

4 Rhinology

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4.0 Anatomy and Physiology of the Nose and Paranasal Sinuses

The external nose consists of the nasal pyramid (frontal process of the maxilla), with the paired nasal bones forming the dorsum and meeting the frontal bone superiorly at the glabella. Inferiorly are the upper lateral cartilages and lower lateral (alar) cartilages, which contribute to the nasal tip and nasal valves (**Fig. 4.1**).

The internal nasal and sinus anatomy is complex and variable. The midline nasal septum is composed of the quadrangular cartilage, the perpendicular plate of the ethmoid bone, the vomer bone, and the palatine bone, with an overlying mucosal covering. There are four paired sinuses: the maxillary sinus, frontal sinus, ethmoidal air cells, and sphenoidal sinus (**Fig. 4.2**). The lateral nasal wall consists of the inferior, middle, and superior conchae (turbinates); below each concha is its corresponding meatus. The nasolacrimal duct opens into the inferior meatus. The frontal and maxillary sinuses and anterior ethmoidal air cells drain via the middle meatus. The posterior ethmoidal air cells drain via the superior meatus. The sphenoidal ostia are near the level of the superior meatus on the anterior wall of the sphenoidal sinus. Clinically, the relation of the paranasal sinuses to adjacent anatomic structures is important, as it relates to the potential for the spread of infection or an iatrogenic injury. Specifically, the ethmoid roof may be an extremely thin bone along the lateral lamella of the cribriform plate and may vary in its height considerably; intracranial contents lie superiorly. The lamina papyracea is, as its name implies, a paper-thin sheet of bone that separates the orbit from the ethmoidal air cells.

The sphenoid sinus is bounded by the internal carotid artery, optic nerves, cavernous sinus, and sella turcica; an overriding posterior ethmoidal (Onodi) cell may risk critical structures. Dehiscence of the bone covering the internal carotid artery within the sphenoidal sinus is relatively common and should be routinely assessed for on a preoperative computed tomography

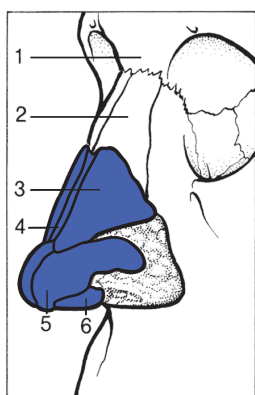


Fig. 4.1 Nasal skeleton. 1. Glabella. 2. Nasal bone. 3. Upper lateral cartilage. 4. Upper edge of the cartilaginous nasal septum. 5. Lower lateral cartilage. 6. Medial crus of lower lateral cartilage. (Used with permission from Becker W, Naumann HH, Pfaltz CR. *Ear, Nose, and Throat Diseases: A Pocket Reference*. 2nd ed. Stuttgart/New York: Thieme; 1994:170.)

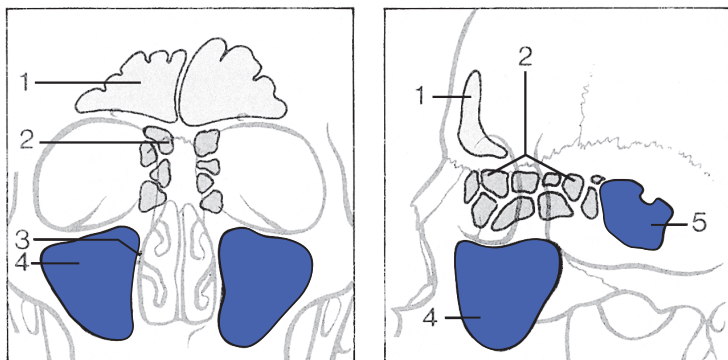


Fig. 4.2 Nasal sinuses: (a) frontal section, (b) sagittal section. 1. Frontal sinus. 2. Ethmoid sinus. 3. Maxillary ostium. 4. Antral cavity. 5. Sphenoid cavity. (Used with permission from Becker W, Naumann HH, Pfaltz CR. *Ear, Nose, and Throat Diseases: A Pocket Reference*. 2nd ed. Stuttgart/New York: Thieme; 1994:175.)

(CT) scan. The frontal sinus is bounded by the orbit and the anterior cranial fossa and may also be a source of spread of rhinogenic infection. A Haller cell is an anterior ethmoidal cell that pneumatizes laterally at the orbital floor and can contribute to maxillary sinus drainage problems. Agar nasi cells are anterior ethmoidal cells that pneumatize superiorly and can contribute to frontal sinus drainage problems.

◆ Blood Supply

There is abundant external and internal carotid supply (**Fig. 4.3**). Kiesselbach's area (also known as Little's area) of the anterior septum provides superficial anastomoses. External carotid branches supply the nose externally (via the facial artery), and the maxillary artery, including the sphenopalatine artery, does so internally. Internal carotid branches are supplied via the ophthalmic artery to the anterior and posterior ethmoidal arteries. Venous drainage occurs via facial veins as well as ophthalmic veins, which have valveless intracranial connections to the cavernous sinus and therefore relate to intracranial hematogenous spread of infection. Epistaxis (nosebleed) is discussed in **Chapter 4.1.5**.

◆ Innervation

General sensory supply is via the first and second divisions of the trigeminal nerve, cranial nerve (CN) V. Importantly, the nasal tip is supplied via CN V₁ (the first, ophthalmic, division of the trigeminal nerve). Thus, if possible herpetic lesions involve the nasal tip, ophthalmologic evaluation is indicated to rule out herpes zoster of the eye. Special sensory supply is via the olfactory (first cranial) nerve. Complex autonomic innervation is supplied to mucosa via the pterygopalatine ganglion, regulating vasomotor tone and secretion. The vidian nerve (nerve of the pterygoid canal) contains preganglionic

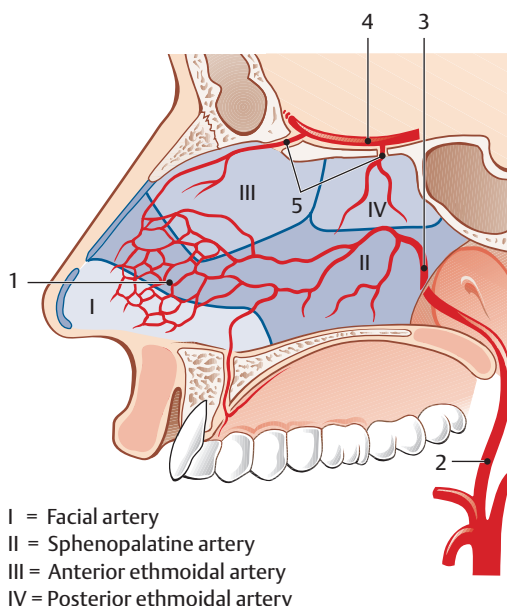


Fig. 4.3 Vasculature of the nasal cavity. 1. Kiesselbach's area. 2. Internal maxillary artery. 3. Sphenopalatine artery. 4. Ophthalmic artery. 5. Anterior and posterior ethmoidal arteries. I–IV: areas supplied by arteries to the nose. (Used with permission from Behrbohm H et al. *Ear, Nose, and Throat Diseases: With Head and Neck Surgery*. 3rd ed. Stuttgart/New York: Thieme; 2009:122.)

parasympathetic fibers from the greater superficial petrosal nerve (from the facial nerve, CN VII) and sympathetic fibers from the deep petrosal nerve (from the superior cervical ganglion) and synapses at the pterygopalatine ganglion prior to innervating the sinonasal mucosa.

◆ Physiology

Table 4.1 gives a summary of tests for the assessment of sinonasal physiology. Warming and humidification of inspired air, olfactory function, and immune function all are aspects of nasal physiology. The nose and paranasal sinuses play a role in host defenses. In general, mucociliary clearance is a key feature. Specific factors such as secretory immunoglobulin A (IgA), lactoferrin, lysozyme, cytokines, and the complex regulation of cells that mediate immunity are critical to the maintenance of normal sinus function. A detailed discussion of sinonasal physiology is beyond the scope of this book. The presence of infection, inflammation, allergy, neoplasm, or traumatic, iatrogenic, or congenital deformity may all perturb sinonasal physiology and must be considered in the evaluation of the patient with complaints related to the nose.

Table 4.1 Summary of tests for factors affecting sinonasal physiology

Lab test	Utility (tests for)
RPR; FTA-ABS	Syphilis
c-ANCA	Wegener's disease
ACE	Sarcoidosis
ANA	Autoimmune disease
ESR	Inflammation
Cultures of sinus drainage	Specific infectious organism
University of Pennsylvania (SIT)	Olfactory function
RAST (blood testing for allergen-specific IgE)	Allergy
Skin testing (prick test; SET)	Allergy
Acoustic rhinometry; rhinomanometry	Nasal obstruction; not currently widely used outside of research settings
Ciliary beat frequency; ciliary electron microscopy	Assess ciliary function and anatomy, e.g., for Kartagener's syndrome; typically tested on adenoid or mucosal biopsy

Abbreviations: ACE, angiotensin-converting enzyme; ANA, antinuclear antibody test; c-ANCA, cytoplasmic anti-neutrophil cytoplasmic antibody; ESR, erythrocyte sedimentation rate; FTA-ABS, fluorescent treponemal antibody absorbed; IgE, immunoglobulin E; RAST, radioallergosorbent test; RPR, rapid plasma reagin; SET, skin end point titration , SIT, smell identification test.

4.1 Rhinologic Emergencies

4.1.1 Acute Invasive Fungal Rhinosinusitis

◆ Key Features

- A rapidly progressive sinonasal fungal infection can be fatal.
- Acute invasive fungal infections occur almost exclusively in immunocompromised or debilitated patients.
- Successful treatment requires early detection, wide surgical débridement, and correction of the underlying predisposing condition.

In the debilitated patient, certain fungal infections can become angioinvasive with tissue necrosis, cranial nerve involvement, and possible orbital or intracranial extension. The most common organisms are *Mucor* or *Aspergillus* species. High-risk patients include those with neutropenia from any cause (e.g., leukemia, bone marrow transplantation), other oncology patients undergoing chemotherapy, patients undergoing chronic immunosuppressive therapy or corticosteroid use, patients with diabetes mellitus

and diabetic ketoacidosis, and patients with acquired immunodeficiency syndrome (AIDS). Acute invasive fungal rhinosinusitis is a distinct and rapidly aggressive disease process that is distinguished by its fulminant course from other forms of fungal sinusitis, such as mycetoma, allergic fungal rhinosinusitis, or chronic invasive (indolent) fungal rhinosinusitis.

◆ Epidemiology

Also known as rhinocerebral mucormycosis, acute invasive fungal rhinosinusitis occurs in the at-risk populations just described: patients with hematologic malignancies, patients post solid organ or bone marrow transplant, patients with diabetes, patients on chronic steroid therapy, neutropenic patients, and patients with AIDS.

◆ Clinical

Signs and Symptoms

A high index of suspicion in any at-risk patient is required, as early diagnosis improves prognosis. A fever of unknown origin should raise suspicion, as should any new sign or symptom of sinonasal disease. Facial edema, periorbital swelling, pain, or numbness are common findings. However, the leukopenic patient may be unable to mount a febrile response. Other findings may include epistaxis, headache, mental status change, or crusting/eschar at the nares that can be mistaken for dried blood. One should consider unilateral cranial neuropathy, acute visual change, or altered ocular motility in an immunocompromised patient to be acute invasive fungal rhinosinusitis until proven otherwise. A black intranasal eschar on exam is considered pathognomonic for invasive *Mucor*.

Differential Diagnosis

A noninvasive sinonasal infection, such as acute bacterial sinusitis, should be considered. An acute bacterial sinusitis complication, such as orbital cellulitis or intracranial suppurative spread, may present similarly. Radiographically similar processes may include squamous cell carcinoma, sinonasal lymphoma, and Wegener's granulomatosis.

◆ Evaluation

See **Fig. 4.4** for a diagnostic and treatment algorithm.

Physical Exam

The patient suspected to have acute invasive fungal rhinosinusitis should be seen without delay. The head and neck examination should focus on cranial nerve function and should include nasal endoscopy. Avoid tetracaine spray or other topical anesthetics. Insensate mucosa noted during an endoscopic exam is consistent with invasive fungal infection. Dark ulcers or pale, insensate mucosa may appear on the septum, conchae, palate, or nasopharynx. Early infection may appear as pale mucosa; the presence of dark eschar has been considered to be pathognomonic. Signs of cavernous sinus thrombosis

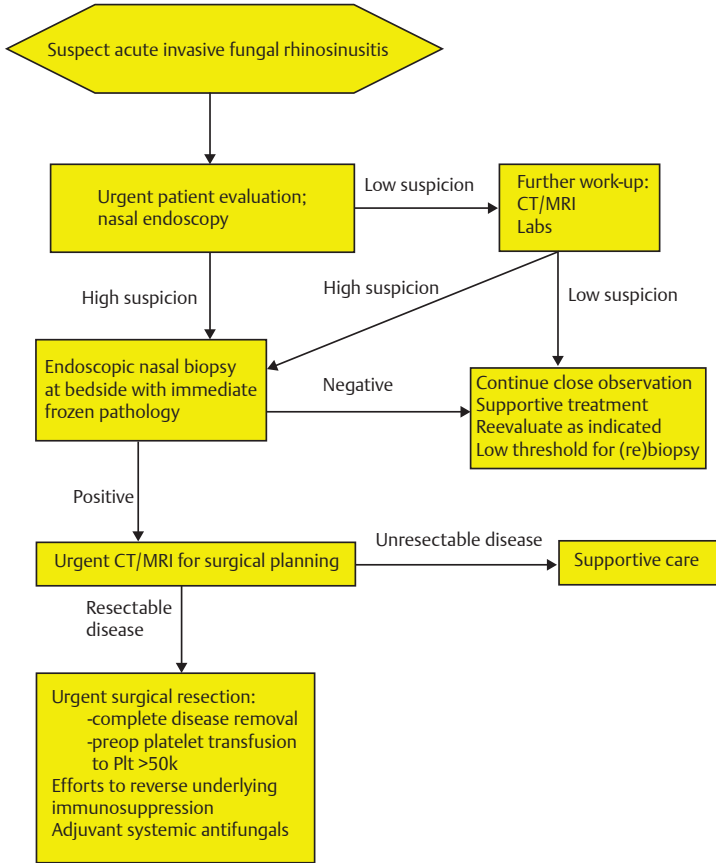


Fig. 4.4 Treatment algorithm for acute invasive fungal rhinosinusitis. Note that there should be a very low threshold to proceed with biopsy, as rapid diagnosis and treatment are critical to patient survival. CT, computed tomography; MRI, magnetic resonance imaging.

include ophthalmoplegia, exophthalmos, and decreased papillary responses. Biopsy of suspicious areas such as the middle concha or septal mucosa is required for diagnosis. It is important to obtain actual tissue at biopsy, not just overlying eschar or necrotic debris. These specimens should be sent fresh for immediate frozen-section analysis as well as silver stain. Patients may be thrombocytopenic, and although a low platelet count may lead to profuse bleeding after biopsy, the risk of this must be balanced with the high mortality associated with a delay in diagnosis. If necessary, platelet transfusion should be ordered early. As a rule, a platelet count of 50,000 is desired. Acceptable hemostasis can usually be obtained with chemical cautery and Avitene (Davol, Inc., Cranston, RI), Gelfoam (Pfizer Pharmaceuticals, New York, NY), or other hemostatic packing.

Imaging

Computed tomography (CT) findings may be nonspecific. However, the presence of bone erosion and adjacent soft tissue edema on contrast-enhanced maxillofacial CT strongly suggests the diagnosis if clinical correlation is present. Unilateral edema of the nasal mucosa has also been associated with invasive fungal sinusitis, as well as obliteration of the retroantral fat planes. Both soft tissue and bone windows, as well as high-resolution axial and coronal views, are necessary. Magnetic resonance imaging (MRI) is useful for evaluating intracranial extension and extension beyond the paranasal sinuses.

Labs

Cultures are inadequate and play no role in the initial diagnosis and management of suspected acute invasive fungal rhinosinusitis. Positive culture results will most likely be available late in the course of the disease. Useful labs for assessing risk factors include a complete blood count (CBC), absolute neutrophil count, chemistries, blood glucose and hemoglobin A1C (HbA1c) in diabetics, and human immunodeficiency virus (HIV) testing with CD4 lymphocyte counts and viral load in AIDS patients.

Pathology

Biopsy of the middle concha or other suspicious lesions with immediate frozen-section analysis is the gold standard test to confirm the presence of tissue-invasive fungus. *Mucor* is identifiable within the mucosa as large, irregularly shaped nonseptate hyphae that branch at right angles. *Aspergillus* is identifiable as smaller hyphae that are septate and that branch at 45° angles. Methenamine silver stain is performed to confirm the diagnosis; however, these results may not be available for several hours.

◆ Treatment Options

This is a surgical emergency: complete surgical resection and the reversal of underlying immune dysfunction are critical. The diabetic patient can be successfully treated with early diagnosis, insulin drip, and wide surgical resection. In the oncology patient, if neutropenia cannot be reversed, mortality is high. Granulocyte-macrophage colony-stimulating factor (GM-CSF) may be useful. Surgical goal is resection of all involved tissue. This may be accomplished endoscopically in select cases. However, an extended total maxillectomy with orbital exenteration may be necessary in advanced disease. Systemic antifungals as well as intranasal nebulized amphotericin are administered, but these should be considered adjuvant therapy.

◆ Outcome and Follow-Up

Prognosis is very poor with intracranial involvement. A bone marrow transplant patient with uncorrectable neutropenia has a poor prognosis. Overall survival in diabetic patients may approach 80% if ketoacidosis is corrected.

4.1.2 Orbital Complications of Sinusitis

◆ **Key Features**

- Sinusitis is a common cause of orbital infection.
- Significant morbidity and even mortality can result.
- Orbital extension of sinusitis is most common in pediatric patients.
- Combined otolaryngology and ophthalmology care is required.

Orbital extension of sinonasal disease requires immediate attention, as rapid progression and blindness may occur. Anatomically, the orbit is bounded by all paranasal sinuses, and infection may spread to the orbit directly or via retrograde thrombophlebitis. The Chandler classification system is heuristically useful in staging and managing orbital complications of sinusitis (**Table 4.2**). Hospital admission and intravenous antibiotic therapy are required for treatment; surgical drainage is necessary in the event of abscess formation, vision compromise, or lack of improvement with medical therapy.

◆ **Epidemiology**

Orbital complications occur in ~3% of sinusitis cases. These are most common in children but can occur at any age. Subperiosteal abscess is present in ~20% of cases of orbital extension of sinusitis. Cavernous sinus thrombosis is rare. Immunosuppressed patients are at increased risk and require aggressive treatment.

◆ **Clinical**

Signs and Symptoms

The most common findings are orbital edema, pain, proptosis, and fever. Orbital disease may be the first sign of sinusitis in children. In more advanced cases, there may be gaze restriction and visual acuity change.

Differential Diagnosis

In the pediatric age group, orbital pseudotumor consists of painful proptosis without a fever or leukocytosis. Orbital rhabdomyosarcoma may present with inflammatory changes in 25% of patients. An ethmoid mucocele may

Table 4.2 Chandler Classification

Stage	Description
I	Preseptal cellulitis
II	Orbital cellulitis
III	Subperiosteal abscess
IV	Orbital abscess
V	Cavernous sinus septic thrombosis

present with proptosis, and CT will reveal an expanded sinus in this instance. Other sinonasal causes of proptosis or orbital edema include allergic fungal rhinosinusitis and neoplasm as well as iatrogenic injury. Abnormal thyroid state may cause ophthalmopathy.

◆ Evaluation

Physical Exam

Examination requires the combined input of the otolaryngologist and the ophthalmologist. In general, the patient will have a history of preceding sinusitis or current complaints consistent with acute sinusitis. Fever is usually present. Head and neck exam reveals lid or periorbital edema, erythema, and tenderness. In cases of preseptal (periorbital) cellulitis, the remainder of the eye exam is normal. The presence of proptosis, chemosis, extraocular muscle limitation, diplopia, or decreased visual acuity suggests orbital cellulitis or subperiosteal abscess. With cavernous thrombosis or intracranial extension, findings may include a frozen globe (ophthalmoplegia), papilledema, blindness, meningeal signs, or neurologic deficits secondary to brain abscess or cerebritis. Superior orbital fissure syndrome is a symptom complex consisting of retroorbital pain, paralysis of extraocular muscles, and impairment of first trigeminal branches. This is most often a result of trauma involving fracture at the superior orbital fissure, but dysfunction of these structures can arise secondary to compression. Orbital apex syndrome adds involvement of the optic nerve.

Imaging

Contrast-enhanced computed tomography (CT) scanning with coronal and axial views is the imaging study of choice if there is any suspicion of postseptal involvement (i.e., other than simple periorbital cellulitis). A subperiosteal abscess is identifiable as a lentiform, rim-enhancing hypodense collection in the medial orbit with adjacent sinusitis. The medial rectus is displaced. In the absence of abscess formation, there may be orbital fat stranding, solid enhancing phlegmon, or swollen and enhancing extraocular muscles, consistent with orbital cellulitis. A lid abscess may be present, less commonly. Suspicion of cavernous sinus thrombosis is better evaluated with magnetic resonance imaging (MRI).

Labs

A complete blood count (CBC) with differential may be useful. Preoperative labs should be ordered as indicated.

Pathology

In younger children, microbiology often consists of single aerobic species including alpha *Streptococcus*, *Haemophilus influenzae*, or coagulase-positive *Staphylococcus*. In those over 10 years old, organisms are often mixed and may include anaerobes.

◆ Treatment Options

Medical

All patients should be admitted and treated with serial ophthalmologic exams and intravenous (IV) antibiotics that have good cerebrospinal fluid (CSF) penetration. Generally, a third-generation cephalosporin is used. Oxacillin is often used in children. Older patients are often double-covered with clindamycin for anaerobes. Alternatives include ampicillin/sulbactam, vancomycin, or aztreonam. Antibiotics are adjusted according to cultures, if possible. Systemic steroids are not recommended. Topical nasal vasoconstrictors are useful (i.e., oxymetazoline).

Surgical

A majority of patients with orbital complications require surgery. This is an area of some controversy. Clearly, surgical drainage is required urgently for abscess formation or decreased visual acuity. If the clinical setting allows for close follow-up (i.e., frequent serial ophthalmologic assessment), many clinicians will observe certain cases of preseptal or early postseptal cellulitis. If there is any progression or lack of resolution with medical therapy over 48 hours, surgery is recommended. Surgical drainage may be accomplished endoscopically by experienced surgeons; however, consent for an external ethmoidectomy approach is recommended. Also, the location of certain abscesses may be amenable to external drainage by ophthalmologist. Regardless of approach, the abscess should be drained and the underlying sinus disease should be addressed. For cavernous thrombosis, involved sinuses, including the sphenoid, must be drained; systemic anticoagulation remains controversial.

◆ Outcome and Follow-Up

The natural history of untreated disease (all stages) results in blindness in at least 10%. Most cases of Chandler stage I–IV disease recover well with treatment. There remains up to an 80% mortality rate with cavernous sinus involvement, although new literature reports suggest this figure is high.

4.1.3 Intracranial Complications of Sinusitis

◆ Key Features

- Intracranial complications due to sinusitis are a life-threatening emergency.
- Management requires a multidisciplinary approach including neurosurgical consultation.
- Complications include meningitis, dural sinus thrombosis, and intracranial abscess.

Sinusitis may result in spread of infection intracranially. Extension may occur via osteomyelitic bone, trauma, or venous channels. The frontal sinus is commonly the source, although ethmoid or sphenoid sinusitis can lead to intracranial spread. Complications include meningitis, epidural abscess, subdural abscess, parenchymal brain abscess, and cavernous sinus thrombophlebitis.

◆ Epidemiology

Since the advent of antibiotics, the incidence has decreased dramatically. Currently, probably fewer than 1% of sinusitis cases are complicated by intracranial spread of infection.

◆ Clinical

Signs and Symptoms

The patient with meningitis of a rhinologic origin will manifest signs and symptoms typical of bacterial meningitis. These include high fever, photophobia, nausea and emesis, mental status change, nuchal rigidity, and pulse and blood pressure changes. However, epidural or subdural abscess without meningitis may be more subtle. Fever and headache may be present but are nonspecific. Usually focal signs are absent. Meningeal signs may develop gradually. A parenchymal brain abscess of rhinologic origin (frontal lobe abscess) may initially result in few signs or symptoms. However, this may progress from headache to signs of increased intracranial pressure, vomiting, papilledema, confusion, somnolence, bradycardia, and coma. Cavernous sinus thrombophlebitis results in spiking fevers, chills, proptosis, chemosis, decreased visual acuity and blindness, and extraocular muscle paresis. Infection can rapidly spread to the contralateral cavernous sinus via venous communications.

Differential Diagnosis

Meningitis of nonrhinologic origin must be considered. The incidental finding of paranasal sinus disease on imaging does not necessarily signify a causal relationship.

◆ Evaluation

Pathogenesis may involve several mechanisms. Extension may occur via osteomyelitic bone. Also, traumatic bone disruption may allow communication of infected sinus contents with dura, for example, after posterior table fracture of the frontal sinus. Also, infection may propagate via venous channels in bone or retrograde venous circulation to the cavernous sinus. General hematogenous spread is possible, especially in a severely immunocompromised host.

Physical Exam

Complete head and neck exam is required with careful assessment of all cranial nerves. Nasal endoscopy may reveal active sinonasal disease and provide mucopus for culture and sensitivities. A neurologic exam including orientation to

person, place, and date will reveal any focal deficits and serve as a useful baseline to monitor for any deterioration. The presence of ocular findings or neurologic deficit should prompt ophthalmologic and neurosurgical consultations.

Imaging

Both computed tomography (CT) and magnetic resonance imaging (MRI) are usually obtained. Contrast-enhanced CT scanning will provide detail regarding paranasal sinus disease and will reveal evidence of bony dehiscence that may underlie intracranial spread of infection. Intravenous contrast will reveal an enhancing abscess. Gadolinium-enhanced MRI will provide a more sensitive assessment of dural enhancement and intracranial disease but lacks the bony detail necessary for planning sinus surgery.

Lumbar Puncture

Obtain cerebrospinal fluid (CSF) via lumbar puncture (LP) in the patient with possible meningitis, after obtaining a CT scan. There is a risk of acute tonsillar herniation following LP, almost always seen in patients with a noninfectious intracranial process causing a mass effect with localizing signs and papilledema. See **Table 4.3** for interpretation of CSF values. Cell count (tube 3), protein and glucose (tube 2), and Gram stain with culture and sensitivities (tube 1) are ordered. Cultures with sensitivity will guide antibiotic therapy.

Labs

In addition to CSF studies, other useful labs include blood cultures, complete blood count (CBC) with differential, chemistries, prothrombin time (PT), and partial thromboplastin time (PTT).

◆ Treatment Options

Medical

IV high-dose antibiotics are empirically started following LP and adjusted based on cultures. Broad-spectrum agents with good CSF penetration are

Table 4.3 Interpretation of cerebrospinal fluid findings

	Opening pressure (cm H ₂ O)	Appearance	Cells (per mL)	Protein (mg %)	Glucose (% serum)
Normal	7–18	Clear	0 PMNs 0 RBCs 0–5 monocytes	15–45	50
Acute purulent meningitis	Often elevated	Turbid	Few–20k (WBCs mostly PMNs)	100–1,000	< 20
Viral meningitis, encephalitis	Normal	Normal	Few–350 WBCs (mostly monocytes)	40–100	Normal

Abbreviations: PMNs, polymorphonuclear cells; RBCs, red blood cells; WBCs, white blood cells.

utilized, such as ceftriaxone or cefepime. Vancomycin is often added for double coverage and for possible methicillin resistance. Peak and trough levels must be monitored when vancomycin is used. The role of anticoagulation for cavernous thrombophlebitis remains controversial. Anticoagulation may interfere with the ability to perform intracranial pressure monitoring or craniotomy.

Surgical

The underlying sinus infection is drained by either endoscopic or open approach. If an intracranial abscess is present, it is drained by the neurosurgeon in conjunction with sinus drainage.

◆ Outcome and Follow-Up

Close observation in an intensive care unit (ICU) setting with serial neurologic exams is required postoperatively. Intracranial pressure monitoring may be required.

4.1.4 Cerebrospinal Fluid Rhinorrhea

◆ Key Features

- Cerebrospinal fluid (CSF) rhinorrhea may occur due to iatrogenic injury to the ethmoid roof or cribriform during sinus surgery, or it can be present as “spontaneous leak” due to increased intracranial pressure.
- Thin-cut computed tomography (CT) scan and β -2 transferrin assay are the most useful studies for evaluation, along with magnetic resonance imaging (MRI) for certain cases.
- The skull base can be repaired with endoscopic instruments from below or via neurosurgical approach from above.

Anterior skull base disruption can result in drainage of CSF via the nose. The underlying problem may be the result of trauma, iatrogenic injury, or congenital anomaly, or it may arise spontaneously (often seen with obesity). The management of CSF rhinorrhea related to skull base injury from endoscopic sinus surgery is highlighted. CSF leak from the temporal bone may present with rhinorrhea via the middle ear and auditory (eustachian) tube.

◆ Epidemiology

Although the exact incidence of complications is unclear currently, previous estimates suggest a ~0.5% rate of skull base injury with CSF leak following ethmoid surgery. Rising obesity incidence has resulted in increased recognition of “spontaneous” leak.

◆ Clinical

Signs and Symptoms

Patients present with watery rhinorrhea. Often, the drainage can be provoked by leaning the patient forward with the head lowered. Typically, the drainage has a salty or metallic taste. Headache may be reported. Anosmia and nasal congestion may accompany iatrogenic skull base injury with encephalocele.

Differential Diagnosis

The consideration for watery or clear rhinorrhea includes various forms of rhinitis versus CSF drainage. Vasomotor rhinitis typically is elicited by cold temperatures, physical activity, or other specific stimuli such as eating.

◆ Evaluation

History

The approach to the patient typically begins with a thorough history. A source of skull base damage may, of course, be quite obvious if there was a suspected iatrogenic injury during preceding endoscopic ethmoid surgery. One should note if the drainage is unilateral as well as its duration and severity. Associated complaints such as headache, visual disturbance, epistaxis, and anosmia should be noted. Details of any previous sinonasal surgery, neurosurgery, otologic surgery, or trauma are important. Any previous imaging studies of the head should be obtained, if possible. Previous history of meningitis should be sought. General past history, medication list, social and family history are obtained.

Physical Exam

A complete head and neck exam is performed. Cranial nerves (CNs) I to XII should be tested. A standard test for olfactory function such as the Smell Identification Test (SIT; Sensonics, Inc., Haddon Heights, NJ) is recommended, to document olfactory function prior to treatment. Nasal endoscopy is performed. A 4-mm rigid 30° endoscope is ideal. Drainage, masses, edema, purulence, and prior surgical changes are noted. If the injury is iatrogenic, it may be possible to assess the location and size of the skull base defect. Superior-based soft tissue mass should raise suspicion for encephalocele. Having the patient perform the Valsalva maneuver may result in visible enlargement of a meningocele or encephalocele.

Imaging

Identification of the site of leak may be straightforward or may be difficult. For a known iatrogenic injury, thin-cut axial CT with coronal views will typically verify the site of injury along the ethmoid roof or cribriform plate. Scanning with an image-guidance protocol may be helpful, by permitting the use of computer-assisted surgical navigation for endoscopic repair. For other situations, high-resolution CT is useful to study the bony detail of the anterior skull base and sphenoids. A bony dehiscence may be evident. For

a patient presenting with a soft tissue mass, MRI is the study of choice to assess for possible meningocele or encephalocele, or for cerebritis.

Labs

If rhinorrhea fluid can be collected, this should be sent for β -2 transferrin assay. Usually, approximately 1 mL is required; the laboratory may require refrigeration or rapid handling of the specimen. The presence of β -2 transferrin suggests strongly that the fluid is indeed CSF.

Other Tests

See **Table 4.4** for a summary of available studies.

MR Cisternogram

Magnetic resonance (MR) cisternogram with intrathecal gadolinium can be highly useful for difficult-to-diagnose leaks, such as those in obese spontaneous-leak patients (idiopathic intracranial hypertension) who may have multiple sites of subtle skull base dehiscence, or patients with prior skull base surgery.

Radioactive Pledget Scanning

This study can be done to help confirm and localize the side of a leak site, although is rarely performed at present because of the success of thin-cut CT at identifying site of leak.

Intrathecal Fluorescein

The use of a dye to visualize the CSF leakage endoscopically is extremely helpful, both for endoscopic diagnosis and at the time of repair. For the procedure, a lumbar drain is placed and 10 mL of the patient's CSF is collected, after which 0.1 mL of 10% IV fluorescein is diluted into the 10 mL CSF sample. This is slowly reinjected intrathecally over 5 minutes. The fluorescein causes active CSF leakage to appear greenish endoscopically. Intrathecal fluorescein can cause seizures at higher dosage; however, the protocol described here is widely accepted as safe.

Table 4.4 Diagnostic studies for suspected cerebrospinal fluid (CSF) rhinorrhea

Study	Role
β -2 transferrin assay	Confirm that drainage is CSF
CT scan, thin cut	Assess bony detail of anterior skull base/identify site of leak; assess for intracranial hematoma or pneumocephalus following iatrogenic injury
MR cisternogram	Identify site(s) of leak, if difficult to identify
Radioactive pledget scanning	Confirm presence of CSF leak, provide localizing information
Intrathecal fluorescein	Localize site of leak during nasal endoscopy/endoscopic repair
MRI scan	Assess for anterior skull base encephalocele, meningocele, neoplasm

Abbreviations: CT, computed tomography; MR, magnetic resonance; MRI, magnetic resonance imaging.

◆ Treatment Options

Skull base disruption causing CSF rhinorrhea is managed with surgical repair. Repair of acute iatrogenic injury is discussed separately from other cases.

Repair of Acute Iatrogenic Injury

If the ethmoid roof is injured during sinus surgery, it may be possible to repair the injury. If there is extensive injury, severe bleeding, or obvious intradural injury, it is highly recommended that neurosurgical consultation be obtained, if possible. Concomitant injury to the anterior ethmoid artery can occur, so the orbit should be assessed for lid edema, ecchymosis, and proptosis. If this occurs, medial wall decompression and/or lateral canthotomy, IV mannitol, and IV steroids may be required. To repair the skull base defect, good visualization is required. Mucosa adjacent to the leak site is gently reflected away. Several graft materials have been described. Acellular dermis is widely used (e.g., AlloDerm [LifeCell Corp., Branchburg, NJ]); others prefer a bone graft placed intracranially, especially if the defect is larger than ~0.5 cm. For this graft, a portion of turbinate (nasal conchal) bone is ideal. After the acellular dermis or bone is placed on the intracranial side of the defect, fibrin glue (or similar material) and fascia or other soft tissue is layered on the nasal side of the defect, followed by several layers of absorbable packing material such as Gelfoam (Pfizer, Inc., New York, NY). Large defects may benefit from a pedicled mucosal flap. It is helpful if the patient can emerge from anesthesia smoothly, without “bucking” and straining, and without the need for high-pressure bag-mask ventilation following extubation, to minimize chances of causing pneumocephalus. Postoperatively, once the patient is stable, a head CT is recommended to evaluate for intracranial hematoma or pneumocephalus. If the scan reveals hematoma or significant pneumocephalus, neurosurgical consultation is necessary. Otherwise, postoperative management should include head of bed elevation, bed rest for 2 to 3 days, and stool softeners. Most surgeons recommend antibiotic prophylaxis with Ancef (GlaxoSmithKline, Brentford, Middlesex, UK) or clindamycin. The use of a lumbar drain is debated. For a large defect, lumbar drainage to reduce CSF pressure for 3 days can be helpful.

Elective Repair of Other Anterior Skull Base CSF Leaks

The management is individualized. Depending on surgeon experience, many ethmoid or sphenoid sinus leaks can be approached endoscopically. Intrathecal fluorescein used intraoperatively is very helpful; the lumbar drain may be used postoperatively to reduce CSF pressure. In other cases, neurosurgical colleagues may approach the skull base defect from above; a pericranial flap can be used to close the defect.

◆ Outcome and Follow-Up

Repair of anterior skull base CSF leaks has a good success rate. Complications can include repeat leakage, infection including meningitis or abscess, encephalocele, anosmia, postoperative intracranial bleeding, or pneumocephalus.

4.1.5 Epistaxis

◆ Key Features

- Management of epistaxis, as for other emergency situations, should begin with ABCs (airway, breathing, circulation) and follow standard protocols for resuscitation and treatment.
- Address predisposing conditions.
- Accurate localization of bleeding site is required for treatment.

The word “epistaxis” derives from the Greek *epi*, meaning on, and *stazo*, to fall in drops. A nosebleed may present as anterior (bleeding from the nostril), posterior (blood present in the posterior pharynx), or both. Accurate localization of the bleeding site is the key to treatment. It is estimated that > 90% of epistaxis cases arise from the anterior nasal septum at Kiesselbach’s area (**Fig. 4.3**). However, the nasal blood supply involves both the internal and external carotid systems, and brisk bleeding can arise posteriorly. Major vessels include the anterior ethmoidal, posterior ethmoidal, sphenopalatine, greater palatine, and superior labial arteries.

◆ Epidemiology

Most nosebleeds are self-limited, not requiring medical intervention. An estimated 45 million Americans suffer from at least minor epistaxis. In children, nearly all cases are anterior, often due to digital trauma. Over age 40, the incidence of posterior bleeds rises.

◆ Clinical

Signs and Symptoms

Patients will report bleeding from the nares or the mouth. There may be an obvious antecedent nasal trauma, surgery, or foreign body reported. Hematemesis is common. On exam, blood may be fresh or clotted.

Differential Diagnosis

The existence of epistaxis is established on history and exam. A variety of underlying local and systemic conditions should be considered (**Table 4.5**). Most important to exclude is a neoplasm presenting with epistaxis. Consider juvenile nasopharyngeal angiofibroma in any teenage male with a unilateral sinonasal mass and epistaxis.

◆ Evaluation

History

In most cases, the approach to the patient begins with a thorough history. The exception to this rule is the patient with severe active bleeding or

Table 4.5 Causes of epistaxis

Trauma	Maxillofacial fractures Foreign body Nose picking
Neoplasm	Juvenile nasopharyngeal angiofibroma Squamous cell carcinoma Inverted papilloma Mucosal melanoma Other
Systemic disease	Hereditary hemorrhagic telangiectasia (Osler-Weber-Rendu disease) Wegener's granulomatosis Sarcoid Coagulopathies Thrombocytopenia
Drugs	Chemotherapy Warfarin, aspirin, clopidogrel, etc. Intranasal illicit drug use
Infection	Tuberculosis, syphilis, rhinoscleroma, viral Other

hemodynamic instability that must first be corrected. Important historical information should include how long nosebleeds have been occurring, their frequency, whether bleeding is typically left- or right-sided and whether it is typically anterior or posterior, how long nosebleeds last, and whether packing or cauterization has ever been required. Note any previous nasal or sinus surgeries. If recent sinonasal surgery has occurred, obtaining operative notes may be helpful. General information that is relevant includes a prior history of easy bruising or bleeding; a family history of such problems or known bleeding disorder; bleeding problems with previous surgeries or dental work; history of anemia, malignancy, leukemia, lymphoma, or chemotherapy; other systemic illnesses; or recent trauma. A complete, accurate medication list is needed. Specific attention is directed at any recent use of medications that can promote bleeding, such as aspirin, other nonsteroidal anti-inflammatory drugs (NSAIDs), warfarin, or clopidogrel (Plavix; Bristol-Myers Squibb, New York, NY). Also consider vitamins such as vitamin E and other supplements or herbs, many of which can promote bleeding. Social history is needed. Chronic alcohol abuse can be related to coagulation disorders from impaired liver synthetic function as well as malnutrition and vitamin deficiencies; illicit intranasal drug use may be causative. In summary, awareness and treatment of systemic conditions may be required to obtain definitive, effective management of epistaxis.

Physical Exam

Priorities with any severe epistaxis must focus on the ABCs and on resuscitation. The key to effective treatment is localization of the bleeding source. The head and neck exam should therefore focus on this goal. It is essential to have a nasal speculum and headlight, Frazier tip suction, and oxymetazoline

(Afrin; Schering-Plough Healthcare Products Inc., Memphis, TN) or phenylephrine (Neo-Synephrine; Bayer Consumer Health, Morristown, NJ) spray (for decongestion and hemostasis). Topical 2% Pontocaine can be used for intranasal anesthetic, if needed. If topical 4% cocaine is available, this is a very effective decongestant and anesthetic, used sparingly. The availability of a 4-mm rigid nasal telescope (e.g., Karl Storz 7200B; Karl Storz, GmbH, Tuttlingen, Germany) and light source is ideal, especially to localize a posterior bleed. Often, the otolaryngologist will first need to remove improperly placed or ineffective packing materials. Removal of clots with suction will facilitate identification of bleeding sites, using the equipment just described.

Imaging

Profuse bleeding in a maxillofacial trauma patient should be managed with emergent angiography. An injury to the extradural portion of the internal carotid cannot be controlled with nasal packs; most of these injuries probably exsanguinate in the field. Also, in recurrent or difficult-to-treat epistaxis, angiography may be diagnostic (to localize a bleeding source and guide definitive therapy such as ethmoidal artery ligation) or therapeutic (to embolize an external carotid source).

Identification of an intranasal mass on exam should prompt computed tomography (CT) or magnetic resonance imaging (MRI) to evaluate a possible sinonasal neoplasm or inflammatory disease fully, prior to definitive therapy. However, this should not delay acute treatment of active epistaxis.

Labs

Blood work should include hemoglobin, hematocrit, platelet count, prothrombin time (PT) and partial thromboplastin time (PTT), as well as blood type and screening tests. Bear in mind that hemo-concentration in the under-resuscitated patient may yield laboratory results that do not initially reflect the degree of blood loss. Platelet function studies may be helpful, although results may not be rapidly available. Platelet dysfunction is seen with NSAID use and von Willebrand's disease.

◆ Treatment Options

A patient presenting to the emergency department with severe epistaxis or any suspicion of hypovolemia should have IV access, continuous cardiac monitoring, and aggressive fluid resuscitation.

Generalizations

Correction of severe clotting disturbances will facilitate the efficacy of local therapy for epistaxis (i.e., reversal of over-warfarinization with fresh frozen plasma [FFP] or IV vitamin K; platelet transfusion in the thrombocytopenic oncology patient). Spontaneous mucosal bleeding typically occurs in patients with platelet counts < 20,000. Control of hypertension is essential. Patients with underlying renal failure may need DDAVP (desamino-D-arginine vasopressin) to correct coagulopathy from underlying uremic platelet dysfunction. Some forms of von Willebrand's disease also respond to DDAVP or desmopressin.

Specifics

Localize the source, as previously described, using oxymetazoline spray, suction, a nasal speculum, and/or nasal endoscopy. Avoid removable packing unless other methods are ineffective. Most bleeds are anterior and can be treated effectively with topical vasoconstrictors and direct pressure and/or simple silver nitrate cautery sticks. A small piece of Gelfoam (Pfizer, Inc., New York, NY) or Surgicel (Johnson & Johnson, New Brunswick, NJ) placed over the cauterized area is helpful. If bleeding will not respond to this, nasal packing to place pressure over the bleeding site may be required. Gelfoam, Surgicel, with or without topical thrombin, or Floseal (Baxter, Deerfield, IL) are among the absorbable products available that will not require removal but will dissolve with nasal saline administration. Intranasal salt pork is a highly effective packing material.

For refractory bleeding, removable packing is placed. Many types are available, such as Merocel (Medtronic XOMED, Inc., Jacksonville, FL) sponges. Rapid Rhino (Brussels, Belgium) products, which are covered with a procoagulant and contain an inflatable balloon, are also effective. Experience has shown that an effective removable pack should be left in place about 4 days to enable healing prior to removal. Any patient with packing in the nose should be placed on systemic antibiotics with good gram-positive coverage, such as cephalexin or clindamycin, for prophylaxis against toxic shock syndrome. In the rare situation in which properly placed nasal packing results in ongoing hemorrhage via the posterior nares, an Epistat (Medtronic XOMED, Inc., Jacksonville, FL) balloon pack (or similar device) is very effective. This device has a balloon that is inflated in the nasopharynx and a second balloon that provides pressure intranasally, and it effectively tamponades a posterior bleeding vessel.

Other formal posterior packing techniques and devices are available as well. A patient with a formal posterior pack requires hospital admission with continuous pulse oximetry. A patient failing or rebleeding following these maneuvers is usually sent for angiography and possible embolization. If unavailable, and/or the surgeon is confident that the feeding vessel is identifiable, clipping or cauterization of the sphenopalatine artery can be performed endoscopically; ligation of the internal maxillary system can be performed via a Caldwell-Luc approach and takedown of the posterior wall of the maxillary sinus; or anterior or posterior ethmoidal artery ligation can be performed via an external incision. Recall the 24–12–6 mm rule to locate the anterior ethmoidal foramen, posterior ethmoidal foramen, and optic nerve canal along the orbital wall; however, there is substantial variation with these measurements.

Other Treatments

Therapy for the Osler-Weber-Rendu patient is a challenge. One should rule out pulmonary or intracranial vascular malformations, which may be life-threatening, with appropriate imaging. Avoid packing. Laser [Nd:YAG (neodymium:yttrium-aluminum-garnet), argon, KTP (potassium titanyl phosphate)] treatments at 1024 nm have been effective. Septodermoplasty remains an option in severe cases. Treatment with bevacizumab (Avastin, Genentech, South San Francisco, CA), a vascular endothelial growth factor (VEGF) inhibitor, has emerged as a highly effective therapy, although risks must be considered with this medication.



◆ Outcome and Follow-Up

Ongoing management of underlying medical disorders may be preventive. Moisture with nasal saline and humidification is helpful. Avoid blood-thinning medications while healing.

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